

Pipeline Walkthrough: "Draft, Critique, Revise" for Şefim Temiz Project

1. Workflow Diagram & Overview This pipeline is designed to transform raw project notes into publication-ready professional technical reports for an AI-based Personal Protective Equipment detection project. The workflow consists of three distinct steps with clear handoffs:

- **Step 1: Draft:** AI generates an initial professional draft based on raw input notes.
- **Step 2: Critique:** AI switches roles to a Senior AI Engineer and ruthlessly critiques its own draft for technical accuracy, missing hardware constraints, and tone.
- **Step 3: Revise & Format:** AI switches back to a Technical Writer, absorbs the critique, and rewrites the text entirely to produce the final polished output.

2. Tools & Prompts Used

- **Tool:** Google NotebookLM (utilizing strict source-grounding with uploaded project documentation).
- **Prompt 1 (Draft):** "Act as a technical project manager for the 'Şefim Temiz' project. Turn the following raw notes into a professional, well-structured first draft (around 200 words) suitable for a stakeholder presentation. Raw notes: [Input]"
- **Prompt 2 (Critique):** "Switch your role. Act as a strict Senior AI Engineer. Read the draft you just generated and provide a harsh but constructive critique focusing on technical emphasis (Computer Vision, Edge AI), tone, and missing details. Do not rewrite, just provide 3-4 bullet points."
- **Prompt 3 (Revise):** "Switch back to an Expert Technical Writer. Read the original draft and your own critique. Rewrite the text completely to address every single point of your critique. Make it technically precise and ready for publication."

3. The Five Real Runs

- **Run 1: Algorithm & Dataset** System Architecture & Technical Specification: Şefim Temiz Şefim Temiz is an autonomous video analytics framework engineered for real-time Personal Protective Equipment (PPE) detection—specifically tracking gloves, arm sleeves, masks, and hairnets—within hygiene-critical environments such as food processing units and healthcare facilities. Addressing a key gap in computer vision literature, which predominantly targets macro-level construction equipment or static entry turnstiles, our platform provides continuous 7/24 visual auditing during active work shifts. The system architecture consists of three decoupled layers: an RTSP stream ingestion service connected to existing IP camera networks, a custom-trained YOLOv8 object detection inference engine, and a centralized web dashboard. The deep learning pipeline performs fine-grained multi-class classification (Gloved, Bare Arm/Hand, Hairnet/No Hairnet, and Mask/No Mask) on high-density video streams. To maintain high precision under variable lighting, non-standard camera angles, and rapid worker movement, inference decisions are

filtered through a strict 80% confidence threshold combined with post-processing temporal logic. This post-processing layer eliminates false positives and prevents alert redundancy (alert spamming) during sustained non-compliant presence. Upon triggering a validated non-compliance event, the pipeline extracts a frame snapshot, embeds a timestamped log, and pushes the visual evidence to the management panel via automated websockets for real-time administrative intervention. By operating purely as a software integration over existing camera hardware, Şefim Temiz provides a zero-hardware-cost, scalable edge vision solution.

- **Run 2: Edge AI and Hardware System Architecture & Edge AI Processing Section**
 1. **Topology & Stream Ingestion Pipeline** The Şefim Temiz platform utilizes a software-based Edge AI architecture designed for continuous, automated visual hygiene compliance monitoring. Rather than relying on specialized smart cameras or expensive physical barrier hardware, the system integrates directly into existing facility IP camera networks. Local edge processing nodes ingest live Real-Time Streaming Protocol (RTSP) video streams from camera feeds. The local stream management pipeline handles hardware frame decoding and frame normalization prior to batch inference. By conducting all video stream decoding and computer vision processing locally within the facility network, the system completely avoids continuous cloud video streaming, eliminating bandwidth overhead while guaranteeing worker privacy.
 2. **Computer Vision Pipeline & Model Optimization** At the core of the visual inspection engine is a custom-trained YOLOv8 deep learning model optimized for multi-class object detection and classification. The model scans incoming video frames to classify target regions across primary hygiene parameters: gloved vs. bare hands/arms, hairnets (bone), and protective face masks. To prevent false alarms caused by environmental scene noise—such as shifting ambient lighting, extreme camera angles, or rapid worker movement—the algorithm applies targeted filtering logic and requires a strict confidence threshold exceeding 80% before flagging a rule violation.
 3. **Event Logging & Low-Latency Alert Dispatched** When a non-compliance event (e.g., a bare arm or missing hairnet) crosses the >80% probability threshold, the edge application triggers an automated logging event. Instead of transmitting heavy raw video files, the processing node extracts a single timestamped proof screenshot, logs the event date and time to the local database, and dispatches a lightweight metadata payload to the central web management dashboard. This local-first architecture delivers instantaneous, sub-second alerts to quality control managers, providing reliable 24/7 automated audit capabilities without straining network infrastructure.
- **Run 3: Team and Development Process** System Architecture and Computer Vision Pipeline Developed at the İnönü University National Technology Workshop under the advisement of Assoc. Prof. Dr. Ömer Faruk Alçın, 'Şefim Temiz' is a software-centric computer vision platform designed for continuous, automated Personal Protective Equipment (PPE) auditing without requiring costly hardware retrofits. The system architecture directly ingests live RTSP video feeds from existing IP camera infrastructure into a central processing module powered by a custom-trained YOLOv8 object detection network. The computer vision pipeline executes real-time multi-class classification across critical hygiene categories—discriminating between

gloves vs. bare arms/hands, hairnets vs. bare heads, and masks vs. unmasked faces. To prevent false positive notifications caused by transient motion, the detection pipeline enforces a calibrated 80% confidence threshold before registering a compliance breach. When a violation exceeds this confidence gate, the system automatically triggers an evidence-logging workflow: it captures the exact frame, applies a date-and-time stamp, writes the event log to the database, and pushes real-time alerts to a web-based administrative dashboard. Rigorously evaluated in laboratory environment trials simulating shifting ambient illumination, adverse camera inclination, and dynamic movement, the optimized pipeline successfully filters operational noise to deliver continuous 7/24 hygiene verification for food processing and healthcare facilities.

- **Run 4: Competition Success (3rd Place)**

1. **Milestone Recognition** The "Şefim Temiz" project—an AI-driven visual hygiene monitoring and Personal Protective Equipment (PPE) tracking system—secured 3rd place in the Engineering and Basic Sciences category at the 5th R&D Project Market (AR-GE Proje Pazarı) hosted by Kırşehir Ahi Evran University.
2. **Architecture & Computer Vision Pipeline** The platform implements a software-centric computer vision pipeline designed to integrate directly with existing facility IP camera infrastructure, explicitly eliminating the need for physical turnstiles or high-cost hardware modifications. The end-to-end data workflow operates through four structured stages: **Stream Ingestion:** Real-time video streams from active IP cameras are routed directly into the server software processing layer. **Deep Learning Inference:** A custom-trained YOLOv8 object detection model processes incoming frames to perform fine-grained classification of limb and gear status, categorizing targets into specific classes: Gloves/Bare Arms & Hands ("Eldivenli", "Çıplak Kol/El"), Hairnets ("Boneli/Bonesiz Kafa"), and Masks ("Maskeli/Maskesiz İnsan"). **Decision Logic & Evidence Logging:** When the inference engine identifies a non-compliance state (e.g., a bare arm) exceeding an 80% confidence threshold, an automated alert trigger is generated. The exact frame containing the violation is captured, stamped with temporal metadata (date and time), and stored in the database. **Dashboard Alert Dispatch:** The timestamped visual proof and incident log are pushed synchronously to a web-based management interface, granting shift supervisors and quality assurance managers immediate auditability. By substituting manual monitoring with continuous 7/24 automated visual auditing, Şefim Temiz resolves post-entry inspection blind spots in food manufacturing and healthcare environments without imposing hardware overhead.

- **Run 5: Future Industrial Applications** **Technical Roadmap and System Architecture Expansion** Building upon the baseline Şefim Temiz continuous surveillance infrastructure—which processes multi-channel video streams over existing local IP camera networks using custom-trained YOLOv8 object detection models—the operational roadmap introduces edge hardware acceleration, multispectral sensor fusion, domain adaptation, and fault-tolerant enterprise middleware:

1. **Multispectral RGB-Thermal Registration & Sensor Fusion** To overcome visibility degradation in extreme operational environments (e.g., steam,

airborne particulate, or severe lighting fluctuations), the ingestion pipeline incorporates long-wave infrared (LWIR) thermal camera feeds. To resolve spatial and resolution mismatches between standard RGB streams and low-resolution LWIR optics, the preprocessing engine applies extrinsic sensor calibration using homography matrix transformations. A dual-stream feature fusion layer combines spatial RGB data with thermal intensity maps prior to the YOLOv8 feature pyramid network (FPN), ensuring accurate feature extraction under low-visibility conditions.

2. **Edge Compute Footprint & Runtime Optimization** To eliminate cloud bandwidth overhead and guarantee deterministic sub-50ms inference latency, processing is shifted to local edge gateways (e.g., NVIDIA Jetson devices). Models are exported to ONNX format and compiled into TensorRT engine runtimes using FP16/INT8 post-training quantization. Hardware-accelerated GStreamer pipelines process multi-channel video decoding directly on dedicated hardware, maintaining a consistent 30 FPS throughput per stream without central server dependency.
3. **Cross-Domain Transfer Learning & Dataset Strategy** Expanding model scope from controlled indoor environments (detecting small targets like gloves, hairnets, and sleeves) to heavy industrial floors and construction sites (detecting hard hats and high-visibility vests) introduces severe domain shifts and occlusion variations. The model pipeline addresses this via domain-adversarial transfer learning and synthetic data augmentation (simulating complex outdoor lighting and physical obstructions). Class-rebalancing techniques and Focal Loss functions prevent performance degradation on rare or partially occluded target classes.
4. **Enterprise Middleware & Fault-Tolerant Protocol Integration** The operational deployment adopts an ONVIF-compliant microservices architecture. Edge processing nodes execute real-time event logging via WebSocket channels and REST APIs to enterprise safety dashboards. To handle temporary network drops, edge devices maintain a localized append-only database buffer that holds evidence snapshots and telemetry logs, triggering automatic synchronization once network connectivity is re-established.

4. Time Accounting (Honest Estimate)

- **Setup Cost:** 15 minutes to upload project documents into NotebookLM and design the 3-step prompt chain.
- **Execution Time (AI):** Approximately 3 minutes per run (including prompt pasting and generation time). Total for 5 runs: 15 minutes.
- **Estimated Manual Time:** Writing, self-reviewing, and rewriting technical documents of this caliber manually would take approximately 45 minutes per text. Total for 5 texts: 225 minutes (3 hours 45 minutes).
- **Total Time Saved:** Approximately 3 hours and 15 minutes.

5. Failure Points & Required Human Review While the pipeline is highly efficient, it requires human handoffs at specific points:

- **Hardware Specifications:** The AI sometimes hallucinates or generalizes hardware limits. A human engineer must review the final stage to ensure exact camera specifications or processing delays are factually accurate.
- **Over-Critique:** In the "Critique" stage, the AI occasionally demands data that the project hasn't generated yet. The human operator must actively decide whether to ignore that critique or manually inject the missing context before initiating the "Revise" stage.
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